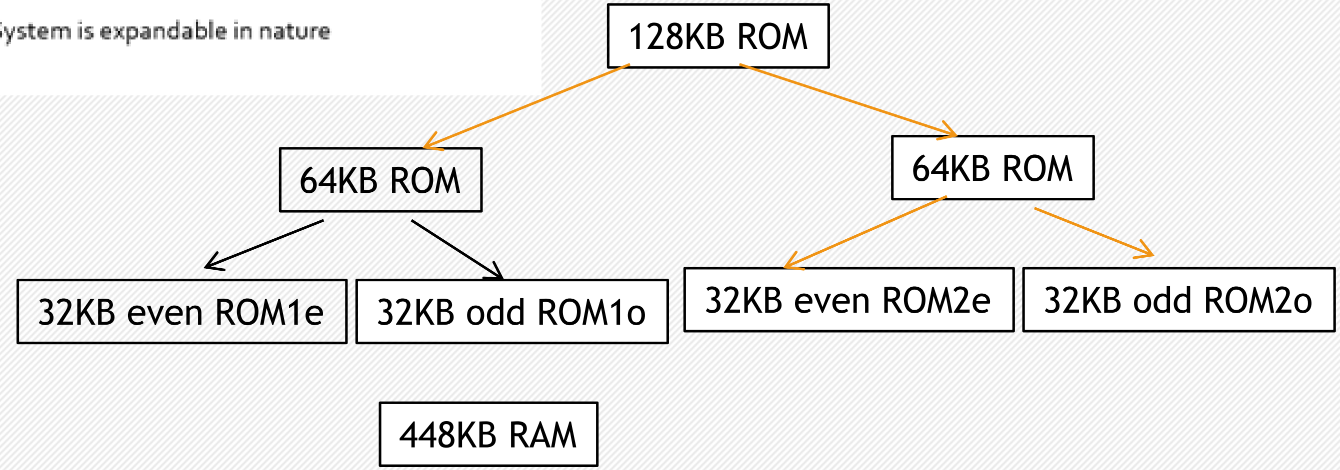
Microprocessor Programming and Interfacing

80286 interfacing

Problem

- 80286-based system has the following memory requirements
- 80286 has 24 bit address bus and 16 bit data bus
- 576KB of memory
- 128KB ROM
- rest RAM
- 64 K ROM 000000H
- 64 K ROM 0F0000H
- RAM 040000H
- System is expandable in nature



7 X 64 =448
So this space is divided into 7 64KB address spaces
Each is further have 32KB even and odd banks
Address begins from 04 00 00H

Chips available:

27256 - 4 (ROM)

61256 - 14 (RAM)

Inverter - 1

LS138 -4

4-i/p OR gates- 1

ROM1:
00 00 00h
to
00 FF FF H

ROM2:
0F 00 00h
to
0F FF FF H

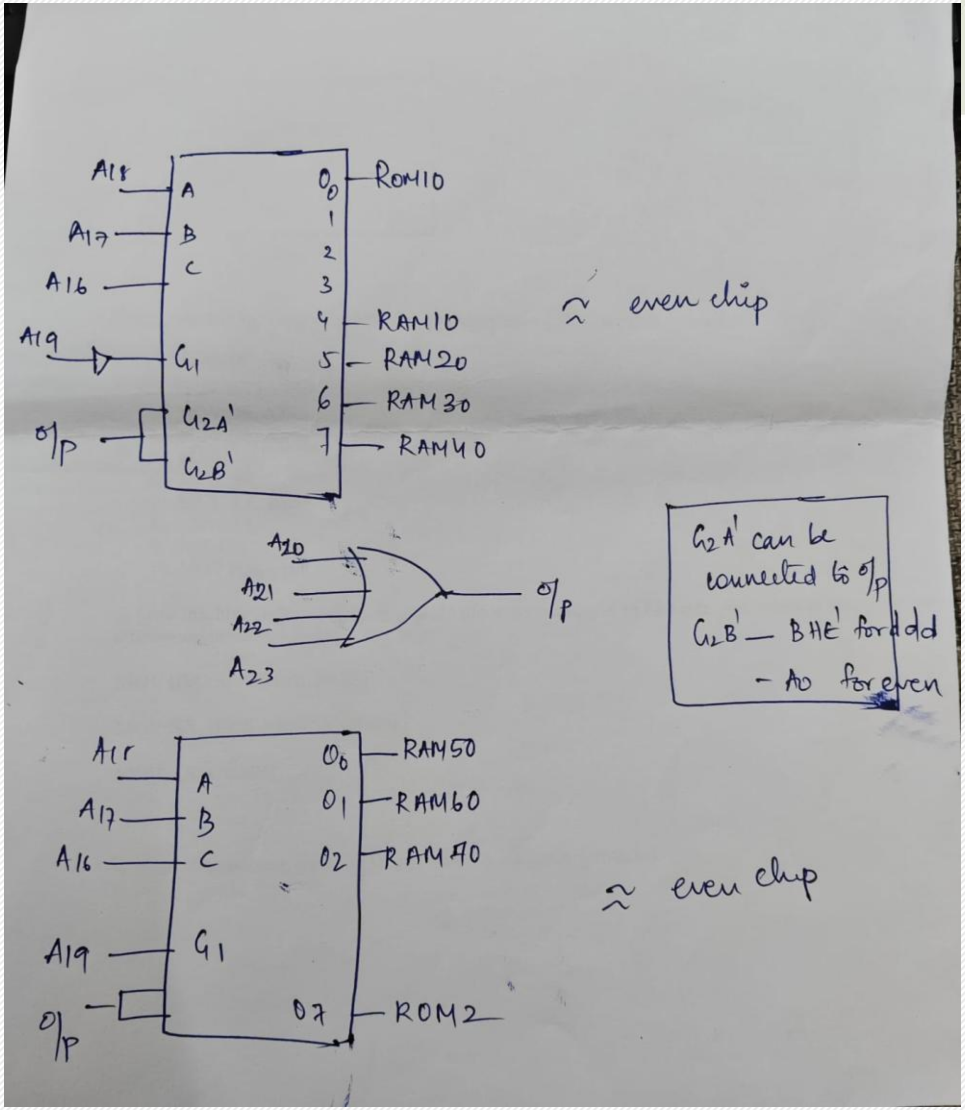
19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

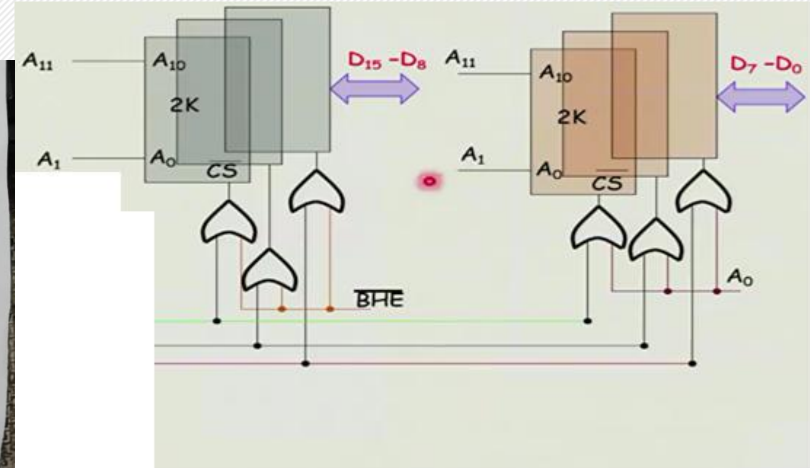
		19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RAM1: 64KB	040000	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	04FFFF	0	1	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
RAM2: 64KB	050000	0	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	05FFFF	0	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
RAM3: 64KB	060000	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	06FFFF	0	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
RAM4: 64KB	070000	0	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	07FFFF	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
RAM5: 64KB	080000	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	08FFFF	1	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
RAM6: 64KB	090000	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	09FFFF	1	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
RAM7: 64KB	0A0000	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	0AFFFF	1	0	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

ROM1:
00 00 00h
to
00 FF FF H

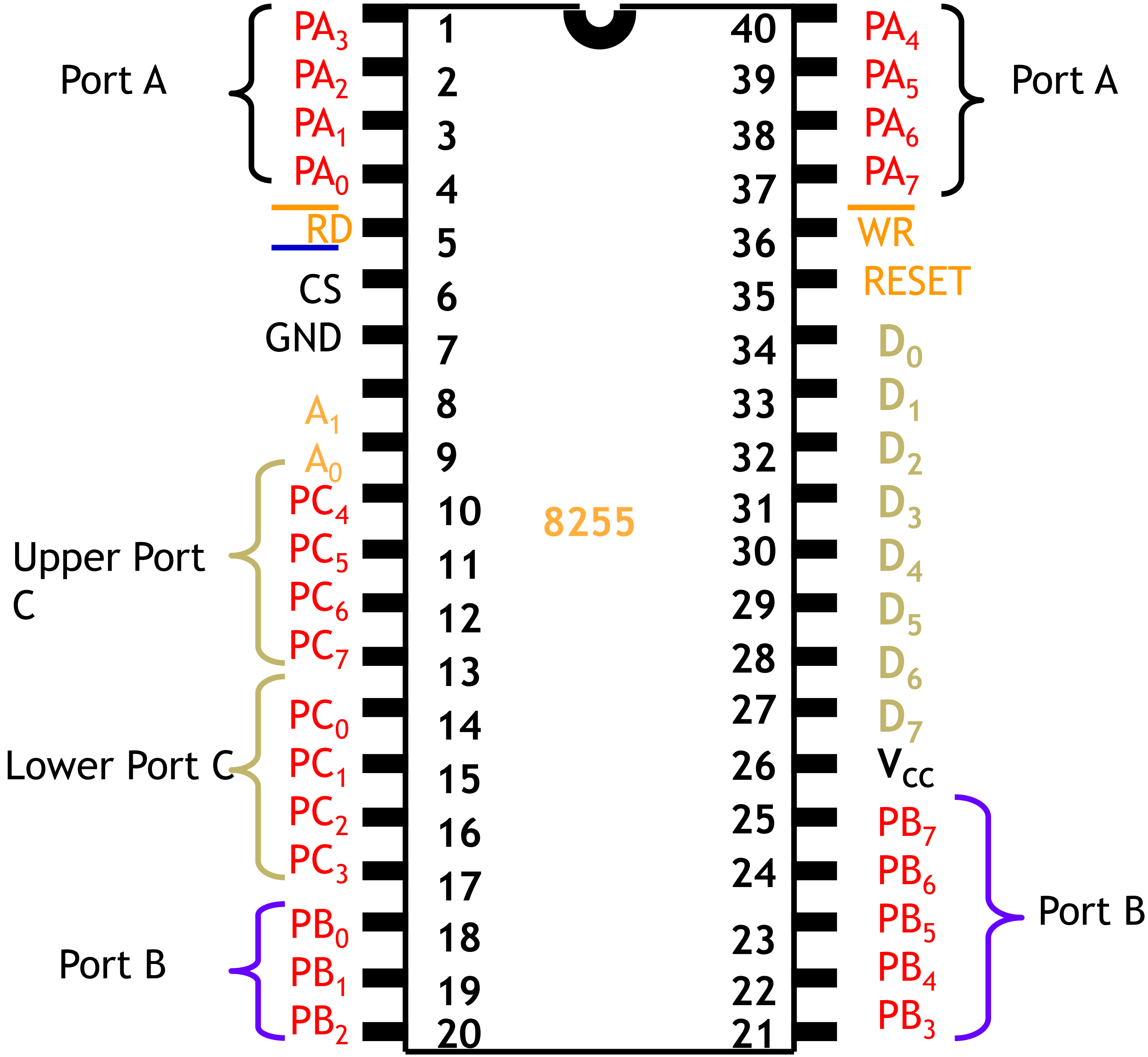
ROM2:
0F 00 00h
to
0F FF FF H

19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

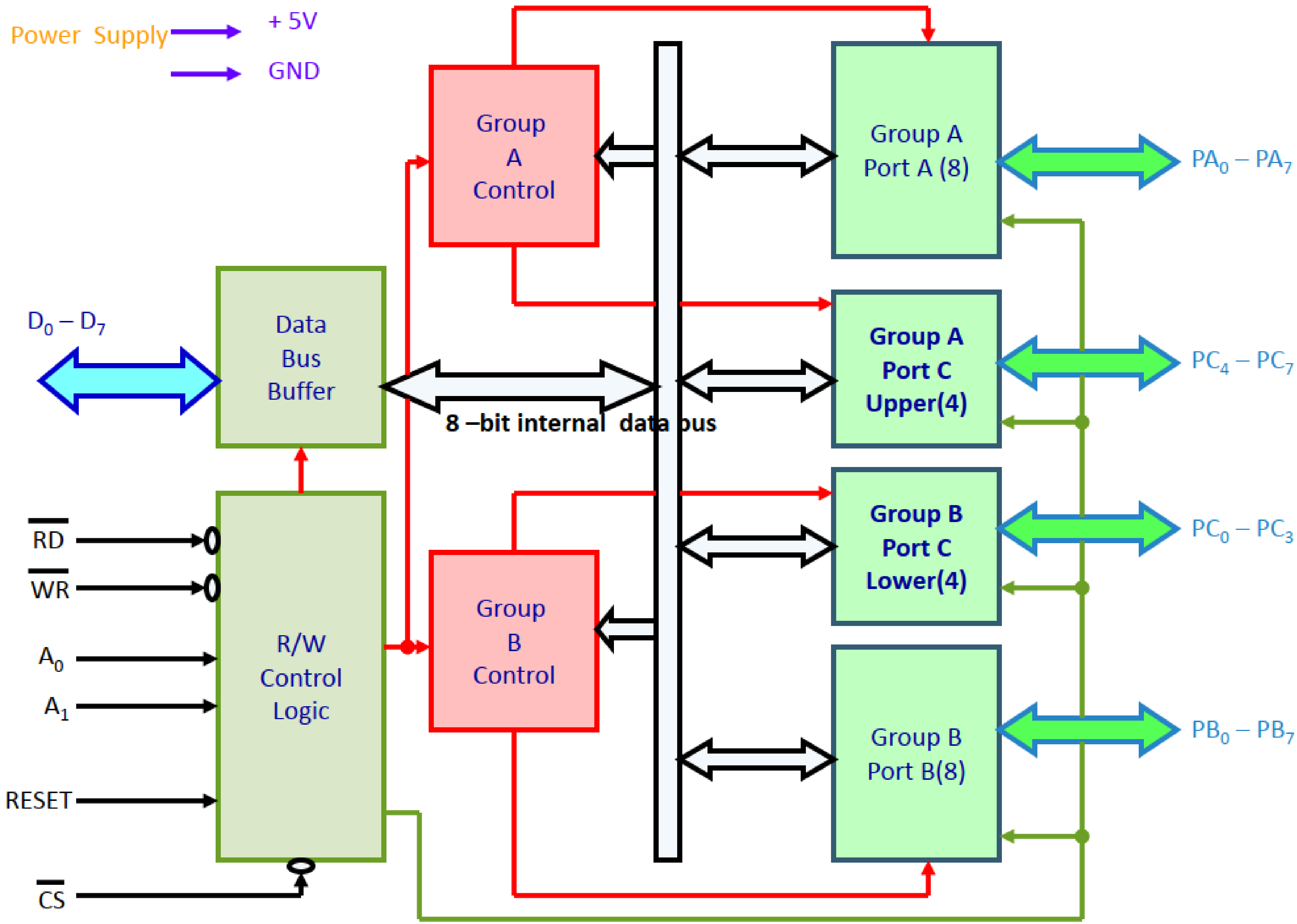

A18
A17
A16
A19
o/p
A
B
C
G1
G2A'
G2B'
O0
1
2
3
4
5
6
7
ROM1O
RAM1O
RAM2O
RAM3O
RAM4O
≃ even chip
A20
A21
A22
A23
o/p
G2A' can be connected to o/p
G2B' — BHE' for odd
— A0 for even
A18
A17
A16
A19
o/p
A
B
C
G1
O0
O1
O2
O7
RAM5O
RAM6O
RAM7O
ROM2
≃ even chip


A11
A10
2K
A1
A0
CS
D15 - D8
A11
A10
2K
A1
A0
CS
D7 - D0
BHE
A0

8255



Port A
PA3 PA2 PA1 PA0
RD
CS
GND
A1
A0
Upper Port C
PC4 PC5 PC6 PC7
Lower Port C
PC0 PC1 PC2 PC3
Port B
PB0 PB1 PB2
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
8255
40 39 38 37 36 35 34 33 32 31 30 29 28 27 26 25 24 23 22 21
PA4 PA5 PA6 PA7
Port A
WR
RESET
D0 D1 D2 D3 D4 D5 D6 D7
VCC
PB7 PB6 PB5 PB4 PB3
Port B



8255 Internal Diagram

CS'	A_1	A_0	Selected
0	0	0	Port A
0	0	1	Port B
0	1	0	Port C
0	1	1	Control Register
1	X	X	8255 Not Selected

Modes of Operation

D_7

0 –BSR

Affects only Port C

1 – I/O

Mode 0

Simple I/O

(A,B,C)

Mode 1

Handshake

I/O -- (A,B)

C - Handshake

Mode 2

Handshake

Bi-directional I/O (A)

C – Handshake

B – Mode 0,1

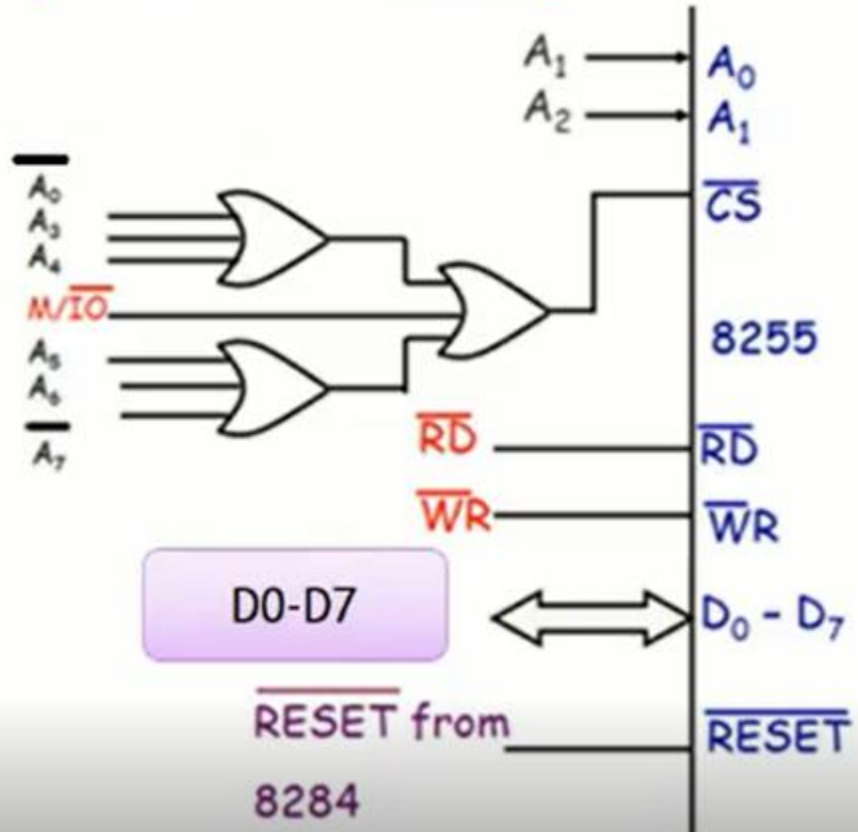
Question 1

- Using 8255 BSR generate a square waveform of frequency of 1 KHz on **PC4.**
- 8255 base address is 100_H.
- Write the software segment for programming 8255 to generate the waveform.
- You can assume that there is a delay routine (DELAY_1) available for generating a delay of 500μs. Write the necessary code snippet to attain the functionality.

what is address of Port A, B , C and CREG ?

100000000 - 100h
100000010 - 102h
100000100 - 104h
100000110 - 106h

System Side Interfacing



Use A8-A15 as well in or gate

CS'	A_1	A_0	Selected
0	0	0	Port A
0	0	1	Port B
0	1	0	Port C
0	1	1	Control Register
1	X	X	8255 Not Selected

D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0
1 – I/O Mode	**Port A Mode** 0 0 - Mode 0 0 1 - Mode 1 1 x - Mode2		**Port A** 1 – i/p 0 – o/p	**Port C Upper** 1 – i/p 0 – o/p	**Port B Mode** 0- Mode0 1- Mode1	**Port B** 1 – i/p 0 – o/p	**Port C Lower** 1 – i/p 0 – o/p
	Group A				**Group B**		

D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0
0 -BSR	x	x	x	Bit_2	Bit_1	Bit_0	Bit Set/Reset
	Don't Care Condition						1 - Set 0 - Reset

PC	0	1	2	3	4	5	6	7
B_0	0	1	0	1	0	1	0	1
B_1	0	0	1	1	0	0	1	1
B_2	0	0	0	0	1	1	1	1

Code

80h: 1 00 0 0 0 0 0

D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0
1 - I/O Mode	**Port A Mode** 0 0 - Mode 0 0 1 - Mode 1 1 x - Mode2		**Port A** 1 - i/p 0 - o/p	**Port C Upper** 1 - i/p 0 - o/p	**Port B Mode** 0- Mode0 1- Mode1	**Port B** 1 - i/p 0 - o/p	**Port C Lower** 1 - i/p 0 - o/p
	Group A				**Group B**		

```
PortA EQU 100h
PortB EQU 102h
PortC EQU 104h
CREG EQU 106h

MOV AL, 80H     ; setting CREG
MOV DX, 106H    ; CREG
OUT DX,AL   ; port C is o/p

X1:        MOV AL,08H
           MOV DX, CREG ;
           OUT DX,AL   ; making PC4 low
           CALL DELAY_1
           MOV AL, 09H
           MOV DX, CREG
           OUT DX,AL
           CALL DELAY_1
           JMP X1
```

08h - 0 000 **100** 0
09h - 0 000 **100** 1

D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0
0 -BSR	x	x	x	Bit_2	Bit_1	Bit_0	Bit Set/Reset
	Don't Care Condition						1 - Set 0 - Reset

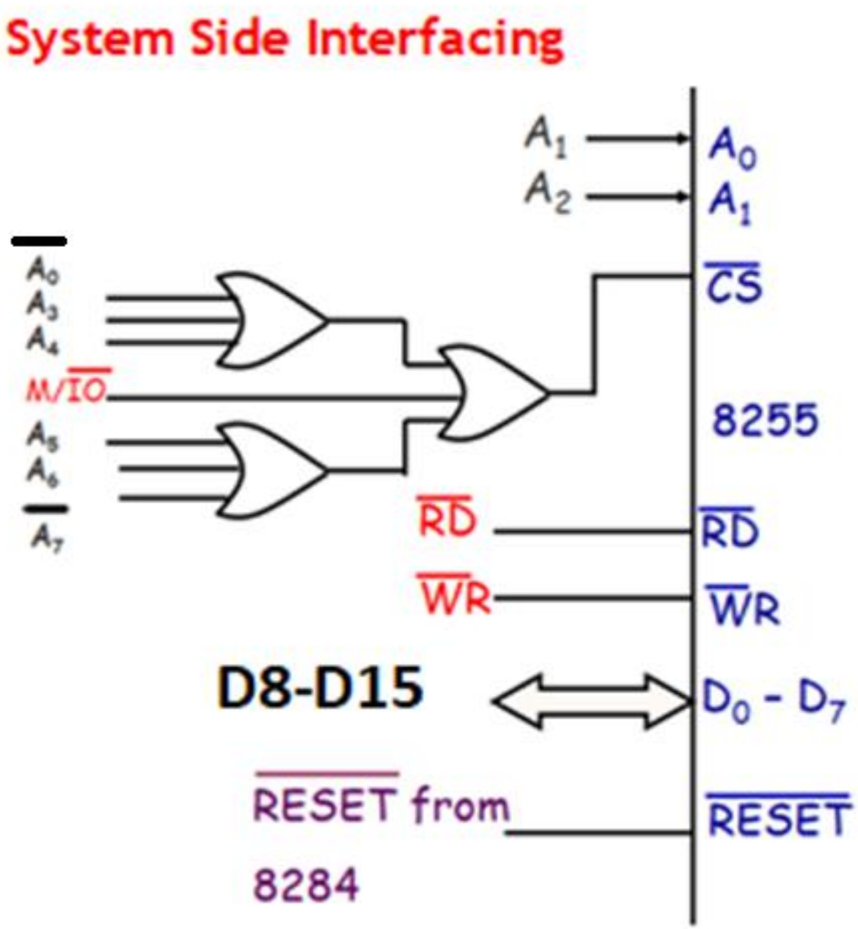
PC	0	1	2	3	4	5	6	7
B_0	0	1	0	1	0	1	0	1
B_1	0	0	1	1	0	0	1	1
B_2	0	0	0	0	1	1	1	1

Question 2

- Suppose that four switches are connected to bits 0 through bits 3 of Port A, an LED is connected to bit 4 of PORT B and another LED is connected to bit 2 of Port C. The switches give a low output when closed (turned on).
- **If odd number of switches are closed**, then LED connected to **port B should be** on and Led connected to Port C should be off.
- **If even number of switches are closed,** then LED connected to port B should be off and **Led connected to Port C should be on**.
- 8255 base address is $\mathbf{81_H}$.
- Show the hardware interfacing (Assuming system bus signals are available) and the necessary code snippets for achieving the functionality.

8255 base address is 81_H. So what is address of Port A, B , C and CREG ?

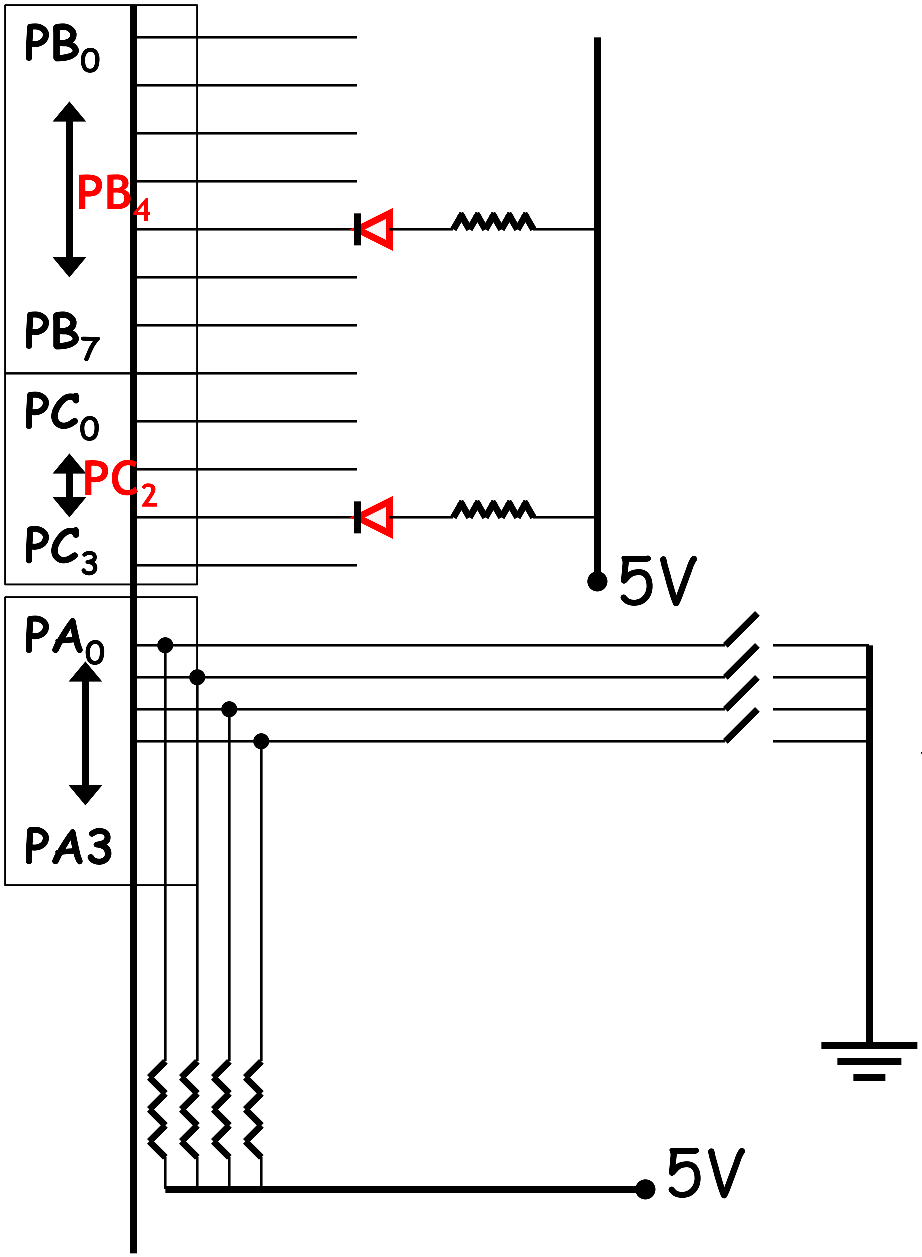
	A7	A6	A5	A4	A3	A2	A1	A0	
Port A:	1	0	0	0	0	0	0	1	- 81h
Port B:	1	0	0	0	0	0	1	1	- 83h
Port C:	1	0	0	0	0	1	0	1	- 85h
CREG :	1	0	0	0	0	1	1	1	- 87h



PortA EQU 81H
PortB EQU 83H
PortC EQU 85H
CREG EQU 87H

CS’	A_1	A_0	Selected
0	0	0	Port A
0	0	1	Port B
0	1	0	Port C
0	1	1	Control Register
1	X	X	8255 Not Selected

Selecting Port / Programming 8255



Suppose that four switches are connected to bits 0 through bits 3 of Port A, an LED is connected to bit 4 of PORT B and another LED is connected to bit 2 of Port C. The switches give a low output when closed (turned on).

8255_1

Interface to the I/O Devices

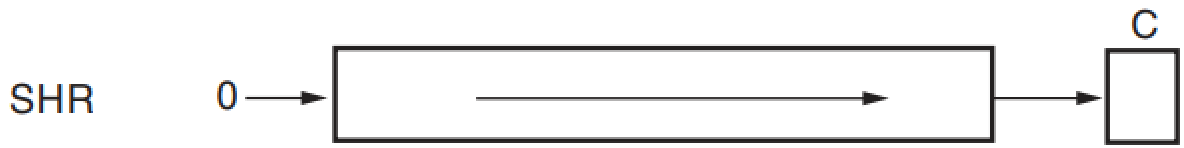
CREG value

Suppose that four switches are connected to bits 0 through bits 3 of Port A, an LED is connected to bit 4 of PORT B and another LED is connected to bit 2 of Port C. The switches give a low output when closed (turned on).

- 98 - 1 00 1 1 0 0 0

D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0
1 – I/O Mode	**Port A Mode** 0 0 - Mode 0 0 1 - Mode 1 1 x - Mode2		**Port A** 1 – i/p 0 – o/p	**Port C Upper** 1 – i/p 0 – o/p	**Port B Mode** 0- Mode0 1- Mode1	**Port B** 1 – i/p 0 – o/p	**Port C Lower** 1 – i/p 0 – o/p
	Group A				**Group B**		

Code




```
PortA EQU 81H
PortB EQU 83H
PortC EQU 85H
CREG EQU 87H
    MOV AL,98H ; port direction config
    OUT CREG, AL
    MOV AH,0  ; counter to store 1's
    MOV CX,4
    IN AL, PortA  ; reading from port A
X2:SHR AL, 1
    JC X1
    INC AH
X1: LOOP X2
    SHR AH,1; checking odd-even
    JNC X3
    MOV AL, 00000000b ; odd
    OUT PortB, AL
    MOV AL, 00000100b
    OUT PortC, AL
    JMP X4
X3: MOV AL, 00010000b  ; even
    OUT PortB, AL
    MOV AL, 00000000b
    OUT PortC, AL
X4:
```

If odd number of switches are closed, then LED connected to **PB4 should be on and Led connected to PC2 should be off.**

If **even number of switches are closed**, then LED connected to **PB4 should be off and Led connected to PC2 should be on.**